

Retail Sales Analysis and Predictive Modeling

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DSC680-T302 Applied Data Science (2265-1)



Business Problem

Goal

Identify the primary drivers of retail sales performance and evaluate whether these factors can be used to support predictive modeling.

Focus

- Analyze key factors influencing retail sales outcomes
- Identify high-performing product categories and geographic regions
- Evaluate the feasibility of predicting sales using available data

This analysis uses the Global Superstore dataset as a retail case study.



Data Overview

Dataset summary

Global Superstore dataset (Kaggle) containing approximately 9,800 retail transactions.

Key variables

Category

Region

Segment

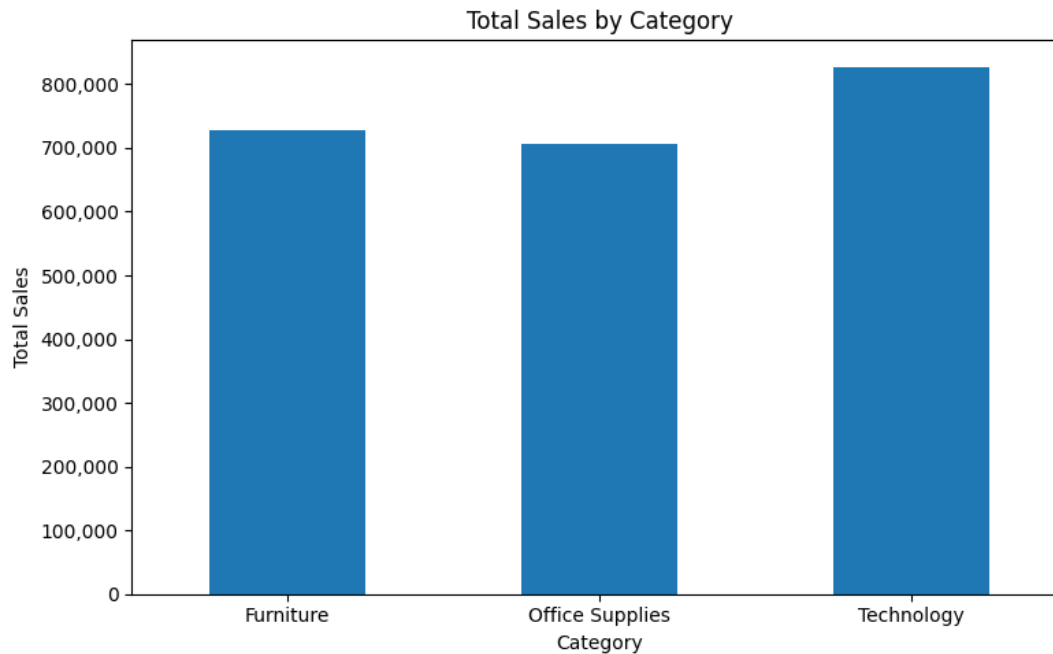
Sales

Order Date

The analysis examines how category, region, segment, and time influence sales performance.



Figure 1. Total Sales by Category



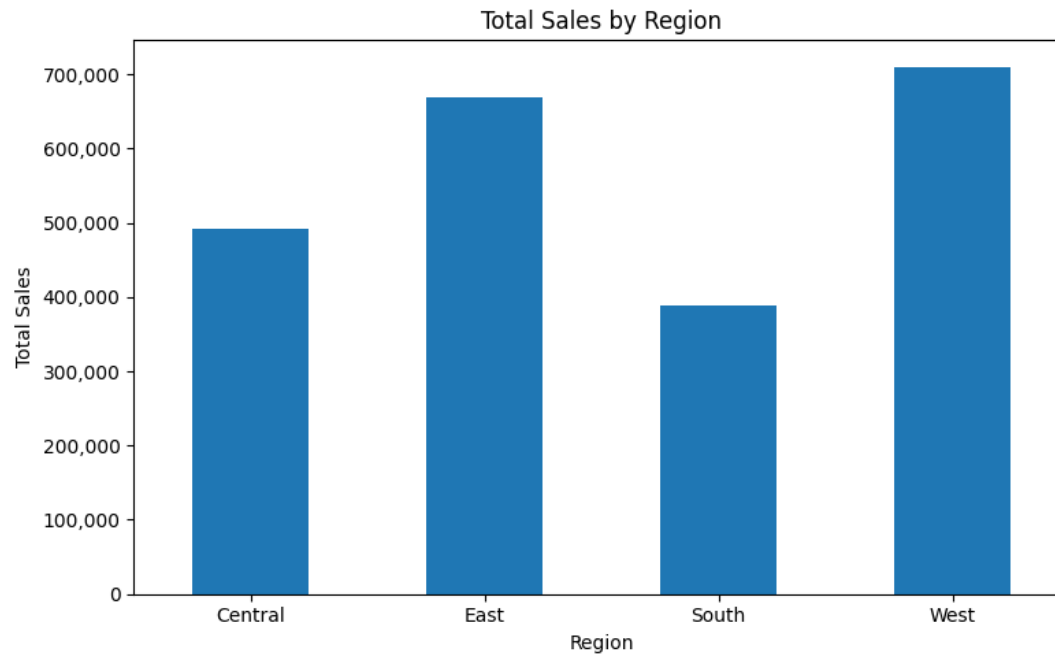
Analysis based on Global Superstore transactions

Key insight

- Technology generates the highest total sales, indicating strong demand and a concentration of revenue in this category
- Furniture and Office Supplies contribute less, suggesting lower relative performance and potential optimization opportunities



Figure 2. Total Sales by Region



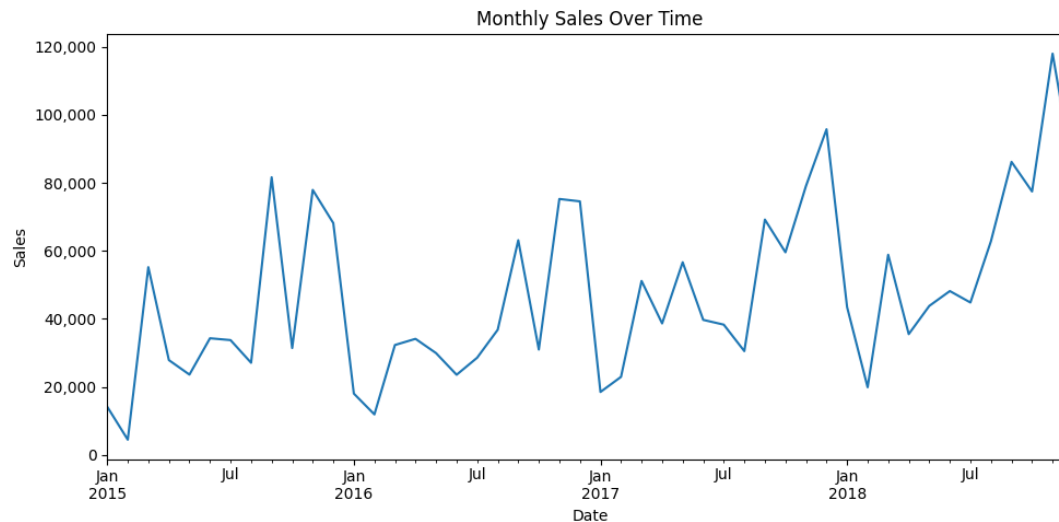
Analysis based on Global Superstore transactions

Key insight

- West and East regions lead in total sales, indicating stronger market performance in these regions
- The South region underperforms, suggesting potential regional inefficiencies or untapped growth opportunities



Figure 3. Monthly Sales Over Time



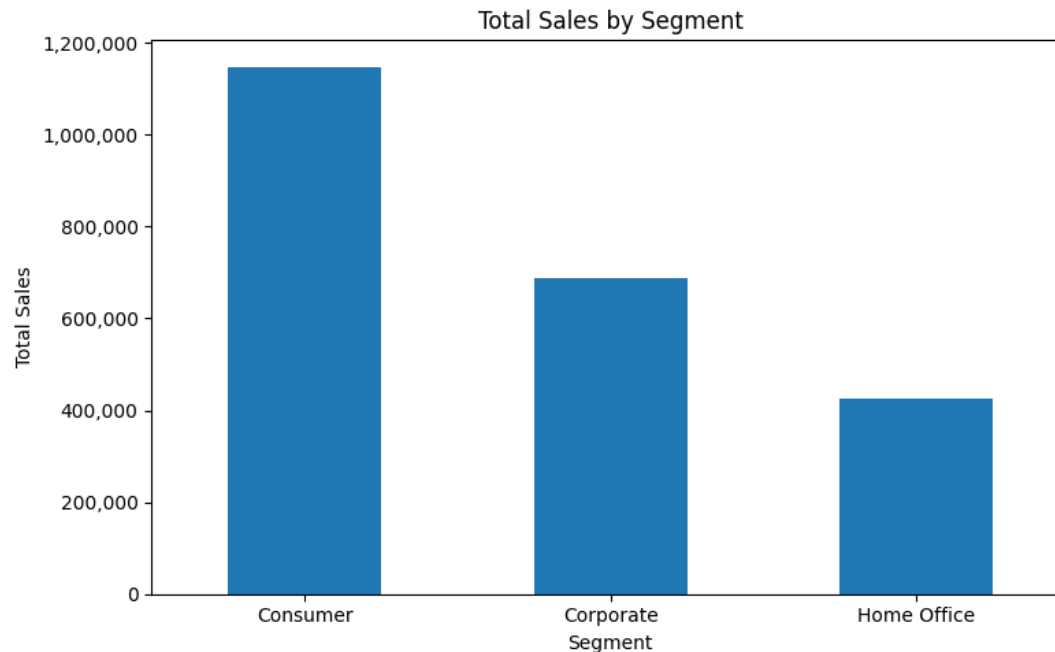
Analysis based on Global Superstore transactions

Key insight

- Sales fluctuate over time, indicating variability in demand across periods
- The overall upward trend suggests sustained growth in sales performance
- Variability suggests potential seasonality or external factors impacting forecasting accuracy



Figure 4. Total Sales by Segment



Analysis based on Global Superstore transactions

Key insight

- The Consumer segment generates the majority of revenue, indicating it is the primary driver of sales
- Corporate sales contribute moderately, while Home Office represents the lowest-performing segment
- This distribution suggests meaningful differences in purchasing behavior and spending capacity across segments



Linear Regression Model Results



$R^2 = 0.04$, indicating the model explains only a small portion of sales variability



Important predictors such as pricing, promotions, and customer behavior are missing



Category and region show the strongest influence on sales



Linear regression is likely too simplistic for complex retail patterns



Key Findings

Sales performance is primarily driven by product category and geographic region

The Consumer segment is the dominant contributor to overall revenue

Sales trends show overall growth but include variability that may impact forecasting

Predictive accuracy is limited due to missing key variables and model simplicity



Recommendations

Prioritize	investment and inventory in high-performing categories such as Technology
Develop	targeted strategies to improve performance in underperforming regions, particularly the South
Expand	data collection to include pricing, promotions, and customer behavior to improve predictive modeling
Explore	more advanced modeling techniques to better capture complex sales patterns



Limitations and Ethical Considerations

Limitations:

- The dataset does not include pricing, promotion, or behavioral variables that significantly impact sales
- The model is limited by the available features and may not fully capture real-world complexity

Ethics:

- The analysis uses non-personal, transactional data only
- Results are interpreted responsibly without overstating predictive capability



Questions?

